

Toy Model I: Deep Linear Networks

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Chapter 2

This chapter was adapted from an interactive blog post on learningmechanics.pub co-authored by Dhruva Karkada, Jamie Simon, and yours truly.

In the previous chapter, we argued that the difficulty of studying deep neural networks comes down to three axes of complexity: coupled dynamics, pointwise nonlinearities, and data complexity. We also learned about the learning mechanic’s go-to approach to tackling these axes: turning one off, and seeing how far the resulting *toy model* can carry us.

In this chapter we turn off the second axis. We keep depth, we keep coupled dynamics, and we maintain a mostly data-agnostic approach, but we throw away the nonlinearities. What remains is a *deep linear network*: a deep neural network with no nonlinear activation functions:

$$\hat{f}(x) = W_L \cdots W_2 W_1 x.$$

In Chapter 1, we solved the training dynamics of the simplest linear model: linear regression (Equation (??)). Doing so revealed that 1) the loss decays exponentially, and 2) out of the infinitely many interpolating solutions, gradient descent picks the one with smallest L2 norm. Those were amusing results, but they didn’t teach us much about the deep neural networks we’re actually interested in because we had thrown away too much complexity. By dropping to a single layer, we killed off the coupled dynamics in addition to the nonlinearities, and coupled dynamics are exactly where a lot of the interesting behavior lives.

This time we keep the coupling. We do so by holding on to depth ($L > 1$), which means several weight matrices influence each other’s gradients, and we ask what that coupling alone—with no nonlinearity—can produce. The answer, remarkably, is *a lot*. The following are four pieces of folklore about how real neural networks train. They’re the kind of empirical findings that the research community might observe and be interested in, but often lack a precise account of *why*:

1. networks preferentially learn certain functional directions before others;¹
2. networks reliably find good (and sometimes globally optimal) minima, despite living in a wildly nonconvex loss landscape;
3. at any given moment in training, each weight matrix’s update is approximately low-rank compared to the matrix itself; and
4. networks sometimes learn in a stepwise fashion—long flat plateaus interrupted by sudden drops in the loss—especially when trained from very small initial weights.

The claim of this chapter is that deep linear networks reproduce *all four* of these behaviors, in a single system we can solve with pen and paper.

¹This is a kind of inductive bias, much like the minimum-norm bias of gradient descent in linear regression from the previous chapter.

1 Setting up the problem

Before we can solve anything, we need to write down the setting precisely. Suppose we have a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ of n sample-target pairs, where both the samples $x \in \mathbb{R}^{d_{\text{in}}}$ and the targets $y \in \mathbb{R}^{d_{\text{out}}}$ are vectors. Our deep linear network passes an input through a chain of matrix multiplications,

$$\hat{f}(x) = W_L \cdots W_2 W_1 x, \quad (1)$$

where the weight matrices $\{W_\ell\}_{\ell=1}^L$ are learned through training so that $\hat{f}(x_i) \approx y_i$. The network is *deep* because it has $L > 1$ trainable layers stacked on top of one another. And it's *linear* because, no matter how many matrices we multiply together, the result is still a single linear map: we can collapse the whole chain into $\hat{f}(x) = W_{\text{tot}} x$, where $W_{\text{tot}} := W_L \cdots W_1$. So as a *function*, this thing is no more expressive than ordinary linear regression. Of course, the interesting part isn't the function it can represent—it's the *path gradient descent takes to get there*, and that path turns out to be genuinely nonlinear.

As usual, we train by minimizing the mean squared error with gradient descent:

$$\mathcal{L}(W_1, \dots, W_L) = \frac{1}{n} \sum_{i=1}^n \|y_i - W_L \cdots W_1 x_i\|^2.$$

Now we'll do a small piece of bookkeeping that pays off enormously later: we'll rewrite this loss so that the data only shows up through a couple of summary matrices, rather than through every individual example. The sum over examples is really an average over the dataset, so we can write the loss as an expectation and do some matrix algebra to simplify the expression:²

$$\begin{aligned} \mathcal{L}(W_1, \dots, W_L) &= \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\text{Tr} (y - W_L \cdots W_1 x)(y - W_L \cdots W_1 x)^\top \right] \\ &= \|\Sigma_{yx} \Sigma_{xx}^{-1/2} - W_L \cdots W_1 \Sigma_{xx}^{1/2}\|_F^2 + \text{const}. \end{aligned} \quad (2)$$

Two new objects appeared here, and they're the only way the data will ever enter our analysis, so let's name them carefully. The first, $\Sigma_{xx} := \mathbb{E}_{\mathcal{D}}[xx^\top]$, is the **input-input covariance**—it records how the input features vary together, with no reference to the targets at all. The second, $\Sigma_{yx} := \mathbb{E}_{\mathcal{D}}[yx^\top]$, is the **output-input covariance** (sometimes the input-output correlation)—it records how inputs and targets co-vary, and so it's where the actual input-to-output relationship we're trying to learn is stored. All of the statistical structure in the data that the network can possibly learn from is packed into these two matrices. Everything downstream depends on the data only through Σ_{xx} and Σ_{yx} .

A note on Equation (2): the $\|\cdot\|_F$ here is the *Frobenius norm*, which just sums the squares of all of a matrix's entries—the matrix version of the ordinary vector length. And the constant term, written out in the footnote,³ doesn't depend on the weights at all. You can think of it as irreducible noise—a floor on the loss that's there no matter what we do with the weights, so we're free to ignore it when we differentiate.

2 Solving the learning dynamics

Like in the linear regression example in Chapter 1, we'll attempt to exactly solve for the learning dynamics of a deep linear network; i.e., write down a formula for what the weights look like at

²Filling in the “some matrix algebra” step—expand the trace and complete the square—is left as an exercise at the end of the chapter.

³const. = $\Sigma_{yy} - \Sigma_{yx}(\Sigma_{xx})^{-1}(\Sigma_{yx})^\top$, which is constant with respect to the weight matrices.

any moment during training. We'll follow a streamlined version of the classic derivation from [Saxe et al. \(2013\)](#). And to keep the algebra easier to follow, we'll only do the $L = 2$ case in full. (The strategy for deeper networks is essentially identical, just more tedious.)

Here's a bird's-eye view of the plan we will execute:

1. **Write the equations of motion.** We'll write down the gradient descent updates, take the infinitesimal-learning-rate limit (gradient flow) from last chapter, and end up with differential equations for how the weight matrices evolve in time.
2. **Make the equations tractable.** We'll look at those equations, decide they're too hard as written, and add one assumption—that the input data is “whitened,” meaning $\Sigma_{xx} = I$.
3. **The hard part: alignment.** We'll argue that when the weights start out very small, the early phase of training mostly does one thing: it rapidly rotates the weight matrices into alignment with each other and with the target, before their magnitudes have grown much.
4. **Decouple and solve.** Once aligned, the matrix dynamics fall apart into independent one-dimensional problems. We integrate those, and out pops a clean sigmoidal learning curve for each “direction” in the data.

2.1 Step 1: Write the gradient flow equations

Our first job is to get from “minimize this loss with gradient descent” to an actual set of differential equations we can try to solve. For a 2-layer network, the loss is

$$\mathcal{L}(W_1, W_2) = \frac{1}{n} \sum_{i=1}^n \|y_i - W_2 W_1 x_i\|^2.$$

To keep the bookkeeping as light as possible, we'll assume everything is square: inputs $x_i \in \mathbb{R}^d$, targets $y_i \in \mathbb{R}^d$, and d hidden neurons in between, so that W_1 and W_2 are both $d \times d$ (none of the conclusions actually need this; the general rectangular case works the same way with more indices to track).

Plain gradient descent with learning rate η updates the weights in discrete steps:

$$\begin{aligned} W_1^{(t+1)} &= W_1^{(t)} - \eta \nabla_{W_1} \mathcal{L}|_{W_1^{(t)}} \\ W_2^{(t+1)} &= W_2^{(t)} - \eta \nabla_{W_2} \mathcal{L}|_{W_2^{(t)}}. \end{aligned}$$

Analyzing discrete steps is pretty much intractable. So we'll turn to a tool introduced in the previous chapter: gradient flow, i.e., infinitesimally small learning rate. This replaces the jumpy update rule with a smooth differential equation in continuous time:

$$\frac{d}{dt} W_\ell = -\nabla_{W_\ell} \mathcal{L}.$$

Now we just need the gradients. Computing them in terms of the two covariance matrices we defined above gives our **equations of motion** for the network:

$$\frac{d}{dt} W_1 = W_2^\top (\Sigma_{yx} - W_2 W_1 \Sigma_{xx}) \tag{3}$$

$$\frac{d}{dt} W_2 = (\Sigma_{yx} - W_2 W_1 \Sigma_{xx}) W_1^\top. \tag{4}$$

Before we try to solve these, let's just read them and get a feel for what they're saying—this kind of “what is this equation telling me physically” pause is always worth taking.

Look at the parenthesized term, $\Sigma_{yx} - W_2 W_1 \Sigma_{xx}$. This is the engine driving both updates; essentially a *residual* term, i.e., the gap between the actual structure in the data (Σ_{yx}) and the structure the model has currently learned ($W_2 W_1 \Sigma_{xx}$). Assuming small initialization, early in training, when the weights are still tiny, the product $W_2 W_1 \Sigma_{xx}$ is negligible compared to Σ_{yx} , so the whole update is dominated by Σ_{yx} —the input-output relationship. In other words, *early learning is steered by the data's input-output structure*, with the input covariance Σ_{xx} playing only a supporting role, gently reshaping the gradient⁴. And what about the end of training? Training stops when the weights stop moving, i.e., when $\frac{d}{dt} W_1 = \frac{d}{dt} W_2 = 0$. Reading off the equations, that happens when $W_2 W_1 = \Sigma_{yx} \Sigma_{xx}^{-1}$.

There's another interesting observation we can make from these equations. Notice how similar they look. If you right-multiply the first by $W_1^\top$ and left-multiply the second by $W_2^\top$, you find that both equal the same thing:

$$\left(\frac{d}{dt} W_1\right) W_1^\top = W_2^\top \left(\frac{d}{dt} W_2\right).$$

Rearranging, this says that a particular combination of the weights never changes during training:

$$\frac{d}{dt} (W_2^\top W_2 - W_1 W_1^\top) = 0. \quad (5)$$

A quantity that stays fixed while everything around it evolves is called a **conservation law**—the same idea as conservation of energy or momentum in physics, where some combination of the moving parts stays pinned no matter how violently the system thrashes around. Here the conserved quantity is the matrix

$$H := W_2^\top W_2 - W_1 W_1^\top,$$

which measures the imbalance between the two layers. When $H \approx 0$, the two layers sit in a kind of equilibrium with each other, and we say the weights are **balanced**.

The algebraic derivation of the conservation law above actually obscures a deeper mechanism. The function that a deep linear network defines has a *continuous symmetry*: it doesn't change if we shrink one layer while correspondingly stretching the next, $W_2 W_1 = (W_2 A^{-1})(A W_1)$ for any invertible matrix A . This symmetry is baked into the fact that we built our model by multiplying matrices together. And it's the "true" source of the conservation law. Physicists will recognize the symmetry-implies-conservation pattern from Noether's theorem. While Noether's theorem doesn't directly apply here, the symmetry really does produce the conserved quantity directly⁵. The conservation law is, at its root, the architecture's symmetry made visible in the dynamics.

An interesting insight we can gain from the conservation law is that the geometry of optimization is highly constrained. Because H can never change, the trajectory the weights trace out is confined to a surface of constant H , which turns out to be hyperbolic in character.⁶ Note that a similar conservation law holds for arbitrarily deep linear networks. Thus, hyperbolic trajectories are a fundamental phenomenon of optimizing linear models using gradient descent.

The conservation law, like any conservation law in physics, allows us to reduce the degrees of freedom in the system. Concretely, the fact that H is pinned for all time means that if we initialize the weights small (so H starts tiny), we get to *assume* the layers stay balanced throughout training (i.e., H stays tiny). This fact will become useful later.

⁴This is kind of matrix that rescales the gradient before it's applied is called a *preconditioner*.

⁵See exercise at the end of the chapter.

⁶For intuition, the scalar version $\frac{d}{dt}(x^2 - y^2) = 0$ says $x^2 - y^2$ is constant; which is exactly the equation of a hyperbola.

2.2 Step 2: Whiten the input data

We now have our equations of motion, but integrating them at this point doesn't seem promising. The trouble is that the matrices Σ_{yx} and Σ_{xx} are tangled together inside the dynamics. These two matrices generally won't *commute*. Since each matrix has its own natural set of axes (its singular vectors), we end up with two conflicting sets of natural axes.⁷ The dynamics are being pulled toward two misaligned coordinate systems at once, and that tension is exactly what makes the equations so hard to integrate.

The most straightforward resolution is to force commutativity. We achieve this by *whitening* the input data: we apply a fixed linear transformation to the inputs that makes their covariance equal to the identity, $\Sigma_{xx} = I$. Whitening just means “decorrelate and rescale the input features so they have unit variance and no cross-correlations”—and crucially, you can always do this as a preprocessing step before training ever begins.⁸ Our equations of motion then collapse to

$$\frac{d}{dt}W_1 = W_2^\top(\Sigma_{yx} - W_2W_1) \quad (6)$$

$$\frac{d}{dt}W_2 = (\Sigma_{yx} - W_2W_1)W_1^\top. \quad (7)$$

Much cleaner. Now only one data matrix remains in the equations: Σ_{yx} .

Finding the right basis

A great deal of applied math and physics comes down to a single move: *find the coordinate system in which your problem becomes simple*. The whole point of whitening was to isolate such a coordinate system. After whitening, all the data-dependent structure in our equations lives in the single matrix Σ_{yx} . The right tool to extract a natural set of axes from it is the **singular value decomposition (SVD)**, which factors any matrix into a rotation, a set of nonnegative stretch factors, and another rotation:

$$\Sigma_{yx} := U^\star \Lambda^\star V^{\star\top}.$$

Here $V^\star$ holds the input-side axes, $U^\star$ holds the output-side axes, and $\Lambda^\star$ is diagonal, holding the stretch factors—the *singular values*—that say how strongly each input axis maps to its partner output axis.

Now we rotate our whole problem into this frame. We define new “tilde” versions of the weights, measured in Σ_{yx} 's coordinate system:

$$\Lambda^\star := U^{\star\top}\Sigma_{yx}V^\star, \quad \tilde{W}_1 := W_1V^\star, \quad \tilde{W}_2 := U^{\star\top}W_2.$$

Substituting these in, the equations of motion keep their exact shape but with Σ_{yx} replaced by the diagonal $\Lambda^\star$:

$$\frac{d}{dt}\tilde{W}_1 = \tilde{W}_2^\top(\Lambda^\star - \tilde{W}_2\tilde{W}_1) \quad (8)$$

$$\frac{d}{dt}\tilde{W}_2 = (\Lambda^\star - \tilde{W}_2\tilde{W}_1)\tilde{W}_1^\top. \quad (9)$$

Since we initialize the weights randomly and isotropically—with no preferred direction of their own—rotating the coordinate axes leaves the *distribution* of initial weights statistically unchanged.

⁷Two diagonalizable matrices (for instance, symmetric ones) commute if and only if they share a common eigenbasis—that is, the same set of natural axes.

⁸This is a perfectly legitimate thing to do to your data, so it's not a crazy assumption. Whether it faithfully reflects what happens in real ML pipelines, where people don't usually whiten, is a separate and fair question.

The rotated problem is therefore exactly equivalent to the original one. We’ve paid nothing and gained a target matrix Λ^* that is now diagonal, with real, nonnegative singular values λ_μ^* (nonnegative because Σ_{yx} is positive-semidefinite).

By rotating into the SVD basis of Σ_{yx} , we’ve revealed that the data is not one tangled blob but a stack of independent, orthogonal pieces, which we’ll call **modes**. Each mode μ couples one *input direction* (the μ -th column of V^*) to one *output direction* (the μ -th column of U^*), and the singular value λ_μ^* tells you how prominent that particular input-output association is in the data. As a rule of thumb, large singular values correspond to coarse, dominant structure, and small singular values to fine details—a fact that will become the whole story when we get to the order in which things are learned.

An enlightening example about the SVD

This mode business can feel abstract, so let’s ground it in a concrete example, due to [Saxe et al. \(2019\)](#). This is also a nice chance to do what we always try to do in this course—check that our abstractions correspond to something real and interpretable.

Imagine the network is shown an item i —say, a “Canary”—encoded as a one-hot input vector x_i (a vector that’s all zeros except for a single 1 picking out “Canary”). Its job is to predict a vector of features y_i : “Can Fly,” “Has Wings,” “Is Yellow,” and so on. Over training it sees many such item-feature pairs, and the statistical structure of this little world is summarized, as always, by the output-input correlation matrix $\Sigma_{yx} = \mathbb{E}_{\mathcal{D}}[y_i x_i^T]$.

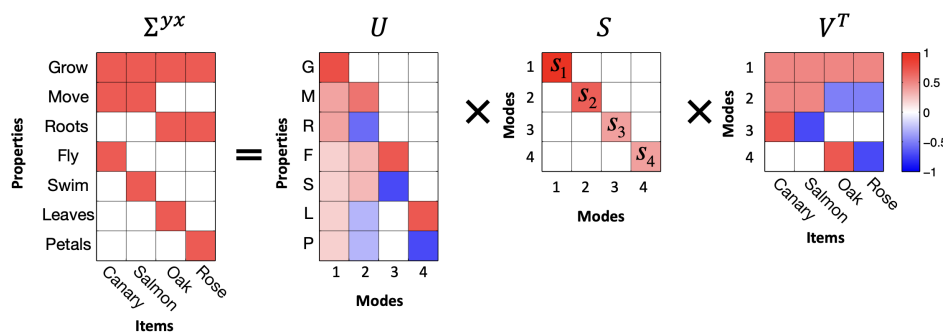


Figure 1: Modes link a set of coherently covarying properties with a set of coherently covarying items.

Now we can give the SVD’s two sets of axes a tangible meaning.

The columns of V^* (equivalently, the rows of $V^{*\top}$) are the **input modes**, or input-analyzing vectors. Each one places every input item along a single interpretable semantic axis. The second input mode, for instance, behaves like an “animal-plant” axis: animals like Canary and Salmon land on the positive side, plants like Oak and Rose on the negative side. A single direction in the input space has captured a coherent semantic distinction.

The columns of U^* are the **output modes**, or output-analyzing vectors. Symmetrically, each one places every output *feature* along a semantic axis. Every input mode is paired with exactly one output mode, and the singular value λ_μ^* joining them measures that axis’s “categorizing power”—how much of the data’s structure this one input-output distinction accounts for. Big singular values are the broad, important cuts through the data (animal vs. plant); small ones are the subtle, subordinate details (which specific kind of bird).

The thing to take away—and the reason this example is worth dwelling on—is that the SVD isn’t just an algebraic convenience we reached for to make an integral tractable. It is carving

the data up along genuinely meaningful semantic directions, and those very directions, as we’re about to see, are the directions the network learns *one at a time, in order of importance*.

2.3 Step 3: Argue that weight alignment happens quickly

Our diagonalized equations of motion are cleaner than before, but they’re still coupled: $\tilde{W}_1$ and $\tilde{W}_2$ both appear inside each other’s updates, so we can’t yet solve for either one on its own. The entire goal of this step is to find a condition under which the coupling falls apart—under which the dynamics *decouple* into a bunch of independent one-dimensional problems we can integrate by hand.

We’re going to proceed by identifying a special condition (we’ll call it *weight alignment*) that, if true, makes everything decouple. Then we’re going to argue—somewhat informally—that small initialization makes this condition come true very quickly, early in training. We will *not* prove this rigorously. Instead we’ll do something that is, in its own way, just as convincing, and we’ll come back to it at the end of the next step.

What condition would decouple the dynamics?

To find the condition we want, it helps to keep thinking spectrally—that is, in terms of the SVD. Recall that the SVD splits any matrix into *directions* (the singular vectors, U and V) and *magnitudes* (the singular values, S). The coupling in our equations lives entirely in the *directions*. If the directional parts of the two weight matrices were already settled and pointing the right way, the only thing left to evolve would be the magnitudes—and magnitudes are just numbers, which decouple easily.

So let’s write each weight matrix in its SVD form, $W_1 = U_1 S_1 V_1^\top$ and $W_2 = U_2 S_2 V_2^\top$, and ask: under what directional conditions does the model $\tilde{W}_2 \tilde{W}_1$ collapse to a clean product of magnitudes, $S_2 S_1$, with all the rotation matrices canceling out? Working it through,⁹ you need three things to line up:

1. $U_2 = U^*$ — the output directions of W_2 match the data’s output modes.
2. $V_1 = V^*$ — the input directions of W_1 match the data’s input modes.
3. $V_2 = U_1$ — the directions where W_1 “hands off” to W_2 match each other.

We’ll call these three conditions together **weight alignment**. Conditions 1 and 2 say the network is aligned with the *data* (its ends point along the data’s modes); condition 3 says the two layers are aligned with *each other* (they interface cleanly, with no rotational mismatch in the middle). When all three hold, only the singular values along the diagonals of S_1 and S_2 are left to evolve, and—as we’ll confirm in Step 4—those evolve independently, mode by mode. That’s the decoupling we’re after.

Why small initialization gets us there

These three conditions are demanding, and there’s no reason a randomly initialized network should satisfy them. So why should we expect alignment to happen at all?

The answer is that it reliably happens, fast, in one specific setting: the **small initialization regime**, where we start all the weights very close to zero so that they have to *grow* substantially over training. This is in contrast to the *large initialization* or “lazy” regime, where weights start

⁹To see it, write the rotated model in terms of the weight SVDs: $\tilde{W}_2 \tilde{W}_1 = (U^{*\top} W_2)(W_1 V^*) = U^{*\top} U_2 S_2 V_2^\top U_1 S_1 V_1^\top V^*$. Now impose the three conditions: $U_2 = U^*$ makes $U^{*\top} U_2 = I$; $V_2 = U_1$ makes $V_2^\top U_1 = I$; and $V_1 = V^*$ makes $V_1^\top V^* = I$. All the rotation matrices collapse to identities and we’re left with $\tilde{W}_2 \tilde{W}_1 = S_2 S_1$, a product of pure magnitudes.

big and barely move from their random starting point—in which case no interesting directional structure ever develops, and none of the phenomena in this chapter appear. We’ll have more to say about that contrast later. Small initialization forces alignment through two separate mechanisms, one for the layers-with-each-other condition and one for the layers-with-data conditions.

We’ll sketch both. Feel free to skim these—the precise arguments lean on a bit of perturbation theory and on power iteration, and the original [Saxe et al. \(2013\)](#) derivation simply *assumed* fast alignment without proving it. The takeaway you actually need is just that *from small initialization, alignment happens quickly*.

Inner alignment (the layers align with each other), via the conservation law. Here we get to spend the conservation law we banked in Step 1. Recall $H := W_2^\top W_2 - W_1 W_1^\top$ is conserved for all time. From small initialization, the weights start tiny, so H starts tiny—and because it’s conserved, it *stays* tiny even after the weights have grown large. That means that once training is underway,

$$W_2(t)^\top W_2(t) = W_1(t) W_1(t)^\top + (\text{a tiny perturbation}).$$

This near-equality of the two layers’ Gram matrices is what we earlier named **balancedness** ($H \approx 0$). And balancedness is enough to pin the layers’ directions together, by way of a classical result called the **Davis–Kahan $\sin \theta$ theorem**. In plain terms, that theorem says: if two symmetric matrices differ only by a small amount, *and* their own eigenvalues are well separated from each other, then they must share essentially the same eigenvectors. Here the two matrices are $W_1 W_1^\top$ and $W_2^\top W_2$, they differ only by the tiny H , and small initialization makes their eigenvalue gaps grow large compared to $\|H\|$ —so the theorem applies and forces their relevant directions to coincide. That’s precisely condition 3, inner alignment ($V_2 = U_1$). (A companion fact, Weyl’s inequality, similarly forces the two layers’ singular *values* to agree—which we’ll use in Step 4.)

Outer alignment (the layers align with the data), via power iteration. The conservation law explains why the two layers snap to each other, but not why they snap to the *data’s* modes. For that, look again at the whitened equation of motion at the very start of training:

$$\frac{d}{dt} W_1 = W_2^\top (\Sigma_{yx} - W_2 W_1).$$

At initialization the weights are minuscule, so $W_2 W_1 \ll \Sigma_{yx}$ and that term is negligible. The update simplifies to

$$\frac{d}{dt} W_1 \approx W_2^\top \Sigma_{yx}.$$

Look at what this is doing: at each instant the weights are getting multiplied by Σ_{yx} again and again. Repeatedly multiplying a vector (or matrix) by a fixed matrix is exactly **power iteration**, the standard algorithm for extracting a matrix’s *dominant* direction—because the largest singular value gets amplified the most with each multiplication, it quickly drowns out the rest. So this early phase acts like power iteration on Σ_{yx} : it aggressively rotates W_1 ’s top directions to point along Σ_{yx} ’s top directions ($V_1 \rightarrow V^*$), which is conditions 1 and 2, outer alignment. By the time the weights have grown enough that the $W_2 W_1$ term stops being negligible, this rotation is already a done deal—the directions are locked, and only magnitudes remain to evolve.

So both mechanisms point the same way: from small initialization, the network rushes into alignment before it does much else. We’ll *assume* alignment holds from here on. We’ll empirically justify this assumption later on as well.

2.4 Step 4: Decouple the dynamics and solve

Now let’s use the alignment assumption to actually solve for the learning curve. The plan: impose alignment to kill the directional coupling, reduce the matrix equations to a single scalar ODE per mode, and integrate that ODE in closed form.

With alignment assumed, our equations of motion involve only the (diagonal) singular-value matrices:

$$\frac{d}{dt}S_1 = S_2(\Lambda^\star - S_2S_1) \quad (10)$$

$$\frac{d}{dt}S_2 = (\Lambda^\star - S_2S_1)S_1. \quad (11)$$

Every matrix here is diagonal. This has two happy consequences. First, it confirms that alignment is *self-sustaining*: there’s no off-diagonal term left to rotate anything, so once the weights are aligned, gradient flow keeps them aligned forever—alignment is a fixed point of the directional dynamics, not a fleeting coincidence. Second, because the matrices are diagonal, the diagonal entries don’t talk to each other at all. The single coupled matrix problem has shattered into many *independent scalar problems*, one per mode. This is the decoupling we’ve been chasing since Step 3.

We can simplify one more notch. Back in Step 3, balancedness (via Weyl’s inequality) told us the two layers’ singular values agree, $S_1 \approx S_2$ throughout training in the small-initialization regime. So we can drop the subscripts and write a single $S(t) := S_1(t) \approx S_2(t)$. The full network’s end-to-end map then has singular values S^2 —one factor of S from each layer. Focusing on a single mode μ and applying the chain rule to its end-to-end singular value s_μ^2 :

$$\begin{aligned} \frac{d}{dt}s_\mu^2 &= 2s_\mu \frac{d}{dt}s_\mu \\ &= 2s_\mu^2(\lambda_\mu^\star - s_\mu^2). \end{aligned}$$

This is something we can solve. It’s a separable ODE: all the s_μ^2 terms go on one side, all the t terms on the other, and we integrate directly.¹⁰ Out comes

$$s_\mu^2(t) = \frac{\lambda_\mu^\star}{1 + \left(\frac{\lambda_\mu^\star}{s_\mu^2(0)} - 1 \right) e^{-2\lambda_\mu^\star t}} \quad (12)$$

Take a breath and appreciate what just happened: we have an exact, closed-form expression for the strength of every mode of the network at every moment of training. We started with a coupled, nonlinear, matrix-valued dynamical system, and through one symmetry-derived assumption we’ve reduced it to a relatively simple scalar formula.

And the formula has a shape that we can recognize. It’s a **logistic** (sigmoid) function of time: each mode’s strength $s_\mu^2(t)$ sits patiently near its tiny initial value $s_\mu^2(0)$ for a while, then rapidly rises, then saturates at its target value $\lambda_\mu^\star$. Learning a mode is not a steady grind—it’s a sudden transition, a switch flipping from “off” to “on.”

¹⁰Carrying out this integration—separate variables, expand $\frac{1}{s^2(\lambda^\star - s^2)}$ by partial fractions, integrate both sides, and solve for $s_\mu^2(t)$ —is a clean, self-contained calculation that lands exactly on the boxed logistic. It’s left as an exercise at the end of the chapter.

Assembling the modes back together, the picture of the whole network is this: a fast initial alignment phase, after which the network keeps the *same directions* as the data (Σ_{yx} ’s singular vectors) for the rest of training, and only the *magnitudes* change, each riding its own sigmoid:

$$W_2(t)W_1(t) \approx U^* S^2(t) V^{*\top} = \sum_{\mu} s_{\mu}^2(t) u_{\mu} v_{\mu}^{\top}.$$

Does any of this actually happen? Let’s check.

Recall that we made an assumption which we technically didn’t prove—that alignment happens quickly from small initialization—and used it to make a clean prediction, the boxed sigmoid above. That prediction is *testable*. We can train an actual 2-layer linear network from small random weights, watch the strength of each mode evolve, and overlay our formula on top of the measured curves. If our unproven assumption were badly wrong, the prediction would miss, and we’d know it.

It doesn’t miss. The analytical sigmoids (red) sit right on top of the measured dynamics of the real network (blue).

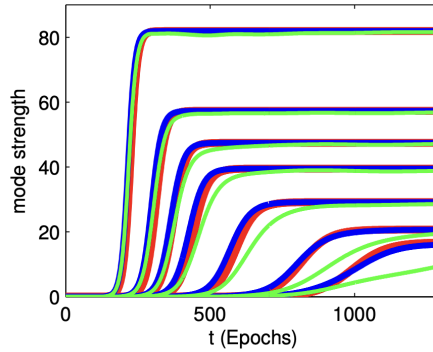


Figure 2: Dynamics of learning in a 2-layer linear network. Curves show the strength of the network’s representation of seven modes of the input-output correlation matrix over the course of learning. Red traces show our analytical sigmoids. Blue traces show simulation of the full dynamics of a 2-layer linear network from small random initial conditions. Green traces show simulation of a *nonlinear* 2-layer network with tanh activation functions.

This illustrates one of the most important tools in a learning mechanic’s toolkit. We rarely get to prove everything rigorously; the systems are too tangled for that. So instead we do this: *make a reasonable-looking assumption, derive a theory from it, and squeeze that theory for concrete, checkable predictions*. Then we go to the empirics. If the predictions come out tight against reality—as they do here—that agreement is itself strong evidence, working backward, that the theory is sound and that the assumption we leaned on was in fact reasonable. A theory that makes sharp predictions and keeps getting them right is a theory worth trusting.

3 Saddle-to-saddle training dynamics

We now have, in the boxed sigmoid from the last section, a complete description of *how* each mode is learned. The natural next question is *when*. If every mode follows its own sigmoid, and each sigmoid switches on at its own time, then the order and spacing of those switch-on times is the whole story of the training curve. Let’s compute it.

A clean way to summarize “when is mode μ learned” is to ask for its **halfway-time** τ_{μ} : the moment at which the mode reaches half its final strength, $s_{\mu}^2(\tau_{\mu}) = \frac{1}{2} \lambda_{\mu}^*$. Setting the sigmoid equal

to $\frac{1}{2}\lambda_\mu^*$ and solving for t (and dropping order-one constants, which don't affect the qualitative story) gives a strikingly simple scaling:¹¹

$$\tau_\mu \approx \frac{1}{\lambda_\mu^*} \ln \frac{\lambda_\mu^*}{s_\mu^2(0)}. \quad (13)$$

Read this formula and a clear prediction jumps out. The learning time is inversely proportional to the mode strength λ_μ^* : **big modes are learned first, small modes last**. And we know from the Canary example what that means semantically—big modes are the coarse, dominant distinctions in the data (animal vs. plant), small modes the fine subordinate details. So the network learns the broad strokes of its task before the fine print, and it does so in a strict, predictable order set entirely by the data's singular values. This is the precise, quantitative version of the first piece of folklore we opened the chapter with.

Now look at the *spacing* between consecutive switch-on times, τ_μ and $\tau_{\mu+1}$. Notice the initialization scale $s_\mu^2(0)$ sits inside a logarithm in the formula: as we shrink the initialization, every τ_μ grows, but they grow by different amounts, and the gaps between them stretch out. The smaller the initialization, the more cleanly separated in time the modes become—the sigmoids stop overlapping and instead fire one after another, in distinct stages.

This is exactly what we see when we plot the loss: a **staircase loss curve**. Training from small initialization, the loss doesn't fall smoothly. It sits on a long flat *plateau*, then drops sharply, then plateaus again, then drops. Our theory tells us precisely what each part of the staircase is: a plateau is a stretch of time when no mode is transitioning (every sigmoid is either still off or already saturated), and a sudden drop is the instant a new mode switches on and the network suddenly captures a new chunk of the data's structure.

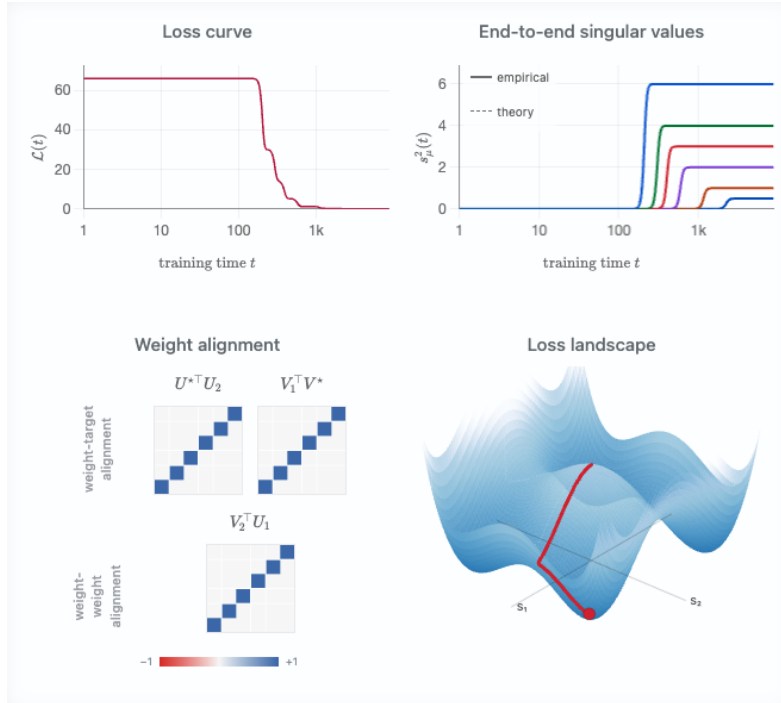


Figure 3: As initialization shrinks, the mode sigmoids separate in time and the loss curve develops distinct plateaus and drops—the staircase.

¹¹Filling in this short calculation is left as an exercise at the end of the chapter.

3.1 Why the staircase? A landscape full of saddles

The right way to understand the staircase loss curve is to picture the geometry of the loss landscape. First we'll show that the origin—where we start, under small initialization—is a saddle point. Then we'll see how each mode contributes its own saddle, so that the full landscape is riddled with them. Finally we'll watch a ball released near the origin thread its way through this landscape, and the staircase will fall right out.

The origin is a saddle

Let's zoom all the way in to the simplest possible case—a single mode, with everything reduced to scalars. The two weights w_1 and w_2 are just numbers, the target is a positive number λ^* , and the loss is

$$\mathcal{L}(w_1, w_2) = \frac{1}{2}(\lambda^* - w_1 w_2)^2.$$

What does this landscape look like right at the origin, $w_1 = w_2 = 0$? Take the gradient and you'll find it's exactly zero there: the origin is a flat, stationary point, a place where the ball, once placed, feels no pull. But “stationary” doesn't tell us its shape. Is it the bottom of a bowl (a minimum), the top of a hill (a maximum), or something else? The way to find out is to stand at the origin and take a small step in a couple of directions and see which way the ground tilts.

Step along the direction where $w_1 = w_2$. Their product $w_1 w_2$ becomes positive, which nudges the prediction $w_1 w_2$ *toward* the target λ^* , so the loss goes **down**—you're walking downhill.

Now step along the direction where $w_1 = -w_2$. Their product becomes negative, dragging the prediction *away* from the positive target λ^* , so the loss goes **up**—you're walking uphill.

A stationary point with a downhill direction *and* an uphill direction is exactly what we mean by a **saddle point**; kinda like the surface of a Pringle. The origin is a saddle.

One saddle per mode, and a combinatorial explosion of them

Now zoom back out to the full, multi-dimensional network. Because the SVD decoupled the dynamics into independent modes, the total loss is essentially just a *sum* of independent single-mode losses (this holds in the aligned, decoupled regime we established in Step 4):

$$\mathcal{L} \approx \sum_{\mu} \frac{1}{2}(\lambda_{\mu}^* - s_{\mu}^2)^2.$$

When is the whole network stationary—when do *all* the gradients vanish at once? Looking at the sum, every mode has to be individually stationary, and we just learned that each mode is stationary in exactly two situations: either it has been *perfectly learned* ($s_{\mu}^2 = \lambda_{\mu}^*$, the bottom of that mode's valley) or it has been *completely ignored* ($s_{\mu} = 0$, that mode's saddle at the origin).

Two possibilities per mode, chosen independently, means that if the data has k modes, the landscape has 2^k stationary points (within this decoupled mode picture). That's a lot of places for a ball to get stuck. Let's sort them:

- Exactly **one** is the global minimum—every mode learned, $s_{\mu}^2 = \lambda_{\mu}^*$ for all μ . This is the answer we want.
- Exactly **one** is the master saddle at the origin—every mode ignored. It's the *highest-loss* of all the critical points (nothing has been learned yet), and it's a saddle, because, as we saw in Movement 1, stepping away from it along any mode's downhill direction lowers the loss.

- **Every one of the other $2^k - 2$ stationary points is also a saddle.** Suppose the largest mode is fully learned ($s_1^2 = \lambda_1^*$) but the second mode is still untouched ($s_2 = 0$). Along the first mode’s coordinate, the network is sitting comfortably at the bottom of a valley. But along the second mode’s coordinate, it is sitting *exactly on that mode’s origin-saddle*, with an unexploited downhill direction right there. A point that is a valley-bottom in some directions and a saddle in others is, overall, still a saddle. So this “first mode learned, rest ignored” state is at an unstable equilibrium.

The kinematics of escaping a saddle

When we initialize with very small random weights, we are setting the ball down a hair’s breadth from the master saddle at the origin. It starts to roll—but because the weights are tiny, the gradients are tiny too, and it rolls *excruciatingly* slowly. This is the source of the very first plateau.

Which way does it go? Recall from Movement 1 that each mode’s downhill steepness is governed by its singular value λ_μ^* —and in fact, near the origin, each mode’s strength grows exponentially at a rate set by λ_μ^* (this is the power-iteration effect from Step 3). So the ball accelerates down the direction of the *largest* mode exponentially faster than any other, slides down the s_1 axis, and escapes the master saddle along that one direction.

Then it arrives at the next saddle: ($s_1^2 = \lambda_1^*$, $s_2 = 0$, $s_3 = 0, \dots$)—first mode learned, rest still at zero, an unstable equilibrium. Because the gradients for every *remaining* mode are still microscopically small, the ball slows down and the loss plateaus. But it hasn’t stopped. Beneath the surface, s_2 is quietly growing, exponentially, driven by the same power-iteration pull. Eventually it grows large enough to tip the ball over the edge of *this* saddle, and the ball slides rapidly down the s_2 axis—which we see, up at the level of the loss curve, as a sudden steep drop. It settles onto the next ledge, and the waiting game begins again for mode 3. Plateau, drop, plateau, drop. This shows up as a staircase loss curve.

It’s remarkable that in this toy example, we’re able to make a concrete connection between what the network is learning and what the optimization dynamics look like. This can be thought of as an instantiation of the learning mechanics idea that learning, at its core, is movement through high dimensional space.

3.2 Saddle-to-saddle as implicit regularization

Notice that because modes are learned in strict order of importance, a network that we stop *partway* through training will have captured only the top few modes of Σ_{yx} , leaving all the rest still pinned near zero. In other words, gradient descent from small initialization has a built-in preference for *simple, low-rank solutions*, adding complexity one mode at a time and only as training proceeds. This is **implicit regularization** again—the very phenomenon we met in Chapter 1, where gradient descent on linear regression quietly preferred the minimum-norm solution without being told to. There, the bias was toward small norm; here, it’s toward low rank. In both cases, gradient descent is doing something we never explicitly asked it to, biasing itself toward simplicity.

4 So what have we learned from all this?

We started with four pieces of deep learning folklore that seemed like clues to deeper understanding which we wanted to capture in a toy model:

1. deep learning learns the components of its target function in a preferred order;

2. deep learning optimizes just fine, despite a wickedly nonconvex loss landscape;
3. instantaneous training dynamics are often low-rank within each weight matrix; and
4. we tend to get stepwise, staircase loss curves, especially when training from small initialization.

Deep linear nets capture all of these. So, what now? Did all this reveal any deeper insight into the fundamental reasons for these folklore observations?

By and large, yes! We can revisit each of these four folklore beliefs with a new perspective.

1. The preferred learning order. We can now say exactly what sets it: the data, through its covariances. The order depends on the input statistics (Σ_{xx}) and the target relationship (Σ_{yx}) together, and the rule is that larger components of the target along larger directions of the input are learned faster—precisely our $\tau_\mu \propto 1/\lambda_\mu^*$ result. Reassuringly, real *nonlinear* networks share this same bias toward quickly learning functions of the data’s large directions which is a sign our linear toy model caught something real rather than an artifact of linearity.

2. Optimizing a nonconvex landscape. We saw that the fearsome nonconvex landscape of a deep linear network actually decomposes, after alignment, into many simple one-dimensional subproblems solved in parallel—one per mode. This decomposition won’t hold exactly for nonlinear networks. But it makes the folklore far less mysterious, and it dovetails with another widely held belief: that large models are full of semi-independent “circuits” that can be understood and optimized more or less separately. A landscape that looks hopeless globally can be benign once you find the right pieces to look at.

3. Low-rank instantaneous dynamics. In our model, the reason is now transparent: at any given moment, typically just one singular value (or a few) is in its rapid-growth phase, while the rest sit still. Each time a singular value switches on, it opens up a new pathway from input to output—it *looks like a new circuit suddenly forming*. It’s possible that deep nonlinear networks show low-rank behavior for the similar reason that only a relatively small number of “circuits” are undergoing rapid development at a particular time. Of course, this is highly speculative and this explanation should be handled with care until tested.

4. Staircase loss curves. This one we explained in full: the parameters slide down a sequence of saddle points of decreasing order on their way to a minimum, and the plateaus and drops are the lingering-and-descending of those saddle-to-saddle dynamics. The same essential mechanism is believed to drive the staircase curves seen in real nonlinear networks, and deep linear networks are now the standard, well-understood toy model for the phenomenon.

So the toy model delivered: four behaviors, one solvable system. But let’s be honest about its limits, in the same spirit we were honest about linear regression at the end of Chapter 1. A deep linear network, no matter how deep, can only ever represent a *linear* function of its input. Every phenomenon that fundamentally requires learning a *nonlinear* function of the data lies completely outside its reach. And that, frustratingly, is *most* of what we ultimately want to understand about deep learning.

Still—and I hope this is the feeling you’re left with—it’s remarkable how far one well-chosen simplification carried us. We kept depth and coupling, gave up the nonlinearity, and in return got an exact, predictive, empirically validated account of preferential learning, benign optimization, low-rank dynamics, and staircase losses. Deep linear networks are the textbook example of a toy

model that is *simple, but not too simple*: simple enough to solve with the tools in hand, rich enough to teach us something true about the real thing.

Chapter 2 Exercises

These exercises fill in the calculations we deferred in the main text. None of them require any tools beyond what's in the chapter.

2.1 The loss in covariance form.

In the setup, we claimed that “with some matrix algebra” the mean squared error can be rewritten purely in terms of the data covariances. Verify this. Starting from

$$\mathcal{L}(W_1, \dots, W_L) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\text{Tr} (y - W_L \cdots W_1 x)(y - W_L \cdots W_1 x)^\top \right],$$

expand the product inside the trace, use linearity of the trace and the expectation, and substitute the definitions $\Sigma_{xx} := \mathbb{E}_{\mathcal{D}}[xx^\top]$, $\Sigma_{yx} := \mathbb{E}_{\mathcal{D}}[yx^\top]$, and $\Sigma_{yy} := \mathbb{E}_{\mathcal{D}}[yy^\top]$ to arrive at

$$\mathcal{L}(W_1, \dots, W_L) = \|\Sigma_{yx}\Sigma_{xx}^{-1/2} - W_L \cdots W_1 \Sigma_{xx}^{1/2}\|_F^2 + \text{const.}$$

where $\text{const.} = \Sigma_{yy} - \Sigma_{yx}(\Sigma_{xx})^{-1}(\Sigma_{yx})^\top$. Confirm that the constant term does not depend on the weight matrices.

Hint: “complete the square” at the level of matrices—add and subtract $\Sigma_{yx}\Sigma_{xx}^{-1/2}$ inside the Frobenius norm.

2.2 The conservation law, from the symmetry.

The algebra trick used in the chapter to derive the conservation law $\frac{d}{dt}[W_2^\top W_2 - W_1 W_1^\top] = 0$ derives it as if by luck. This exercise re-derives it from the continuous symmetry of the loss, making clear *why* the trick works.

Throughout, use the matrix inner product $\langle A, B \rangle := \text{Tr}(A^\top B)$, and recall the gradient flow $\dot{\theta} = -\nabla \mathcal{L}$ for $\theta = (W_1, W_2)$.

- (a) (*Symmetry $\Rightarrow$ orthogonality.*) Suppose $\mathcal{L}$ is invariant under a one-parameter family of transformations $\theta \mapsto \theta + \epsilon v(\theta)$, with generator (velocity field) v . Show that invariance, to first order in ϵ , is equivalent to

$$\langle v(\theta), \nabla \mathcal{L}(\theta) \rangle = 0 \quad \text{for all } \theta.$$

- (b) (*Gradient flow never moves along the symmetry.*) Substitute the gradient flow equation to conclude that $\langle v(\theta), \dot{\theta} \rangle = 0$ along any trajectory. Interpret this geometrically: the gradient flow velocity is always orthogonal to the symmetry direction, so optimization never spends motion traveling along the symmetry orbit.
- (c) (*The specific symmetry.*) For the deep linear network, the symmetry is $W_1 \mapsto A W_1$, $W_2 \mapsto W_2 A^{-1}$. Take $A = e^{\epsilon M} \approx I + \epsilon M$ for an arbitrary generator matrix M , and show that the induced generator is $v = (M W_1, -W_2 M)$.
- (d) (*All generators at once.*) Plug this v into the orthogonality condition from (b), substitute $\nabla_{W_1} \mathcal{L} = -\dot{W}_1$ and $\nabla_{W_2} \mathcal{L} = -\dot{W}_2$, and use the cyclic property of the trace to rewrite both terms with $M^\top$ pulled to the front. Show the condition becomes

$$\text{Tr}(M^\top \dot{W}_1 W_1^\top) = \text{Tr}(M^\top W_2^\top \dot{W}_2).$$

- (e) (*Conclude.*) Since this must hold for *every* generator M , deduce that $\dot{W}_1 W_1^\top = W_2^\top \dot{W}_2$, which is exactly $\frac{d}{dt}(W_2^\top W_2 - W_1 W_1^\top) = 0$. Note where the arbitrariness of M was essential: it is what upgrades a single scalar orthogonality condition into the full matrix-valued conservation law.

Remark. Compare this derivation to the algebra trick. The trick yields $\dot{W}_1 W_1^\top = W_2^\top \dot{W}_2$ by a fortunate cancellation; here the *same* equation appears as the direct consequence of “the flow is orthogonal to every symmetry direction.” The luck of the trick was the symmetry in disguise all along.

2.3 Alignment collapses the model to its magnitudes.

Write the weight SVDs as $W_1 = U_1 S_1 V_1^\top$ and $W_2 = U_2 S_2 V_2^\top$, and recall the rotated model $\tilde{W}_2 \tilde{W}_1 = U^{\star\top} W_2 W_1 V^\star$, where $\Sigma_{yx} = U^\star \Lambda^\star V^{\star\top}$. Show that under the three weight-alignment conditions,

$$U_2 = U^\star, \quad V_1 = V^\star, \quad V_2 = U_1,$$

the rotated model reduces to the product of magnitude matrices, $\tilde{W}_2 \tilde{W}_1 = S_2 S_1$, with all rotation factors canceling. Identify which orthogonality relation kills each rotation factor.

2.4 Solving the scalar learning dynamics.

This is the climactic integration of the chapter. The μ -th end-to-end singular value obeys the separable ODE

$$\frac{d}{dt} s_\mu^2 = 2s_\mu^2(\lambda_\mu^\star - s_\mu^2).$$

Let $u := s_\mu^2$ for brevity, so that $\dot{u} = 2u(\lambda_\mu^\star - u)$.

- (a) Separate variables and use the partial-fraction identity

$$\frac{1}{u(\lambda_\mu^\star - u)} = \frac{1}{\lambda_\mu^\star} \left(\frac{1}{u} + \frac{1}{\lambda_\mu^\star - u} \right)$$

to integrate both sides from 0 to t .

- (b) Solve the resulting expression for $u = s_\mu^2(t)$ and confirm you recover the logistic solution

$$s_\mu^2(t) = \frac{\lambda_\mu^\star}{1 + \left(\frac{\lambda_\mu^\star}{s_\mu^2(0)} - 1 \right) e^{-2\lambda_\mu^\star t}}.$$

- (c) Check the two limits: confirm that $s_\mu^2(t) \rightarrow s_\mu^2(0)$ as $t \rightarrow 0$ and $s_\mu^2(t) \rightarrow \lambda_\mu^\star$ as $t \rightarrow \infty$.

2.5 Time-to-learn.

Using the logistic solution from the previous exercise, derive the halfway-time τ_μ at which mode μ reaches half its final strength, $s_\mu^2(\tau_\mu) = \frac{1}{2} \lambda_\mu^\star$. Show that, dropping order-one constants,

$$\tau_\mu \approx \frac{1}{\lambda_\mu^\star} \ln \frac{\lambda_\mu^\star}{s_\mu^2(0)}.$$

Then argue from this expression that (a) larger modes are learned first, and (b) the time-separation between consecutive modes widens as the initialization scale $s_\mu^2(0)$ shrinks—the mechanism behind the cleanly separated stairsteps in the loss curve.